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SUBSECTION 4.4 ELECTRICAL WORKS

4.4.1. GENERAL

The Bidder shall install, interface with other Bidders where required, perform checking and testing (before, during and after installation), commission and put into successful operation all electrical equipment, components, controls, instruments and accessories. The Bidder shall do all work and furnish all equipment, components, materials, etc. required for a complete system in accordance with this Contract.

The Bidder shall provide all electrical work and materials required for, and incidental to the power plant installation, assembly, mounting, wiring, testing and commissioning for all equipment, components, controls, instruments and appurtenances furnished by the Bidder as specified and as generally described in the **Part 4, Subsection 1.2, Summary of Work**.

The Bidder shall install and test the equipment furnished by him detailed in **Part 4, Subsection 1.2**. The interconnecting cabling systems required for the Work shall be designed, furnished, installed, terminated, tested and commissioned by the Bidder in accordance with Technical Specifications.

The Bidder shall be responsible for complete installation of the wiring and conduits from electrical items; such as, control stations, limit switches, pressure switches, temperature switches, flow switches, sensing elements (such as thermocouples and resistance temperature detectors) and motors to suitable local junction boxes, local control panels or combination starters which shall also be furnished and installed by the Bidder including all local junction boxes, conduit, fittings, cable, wiring, steel supports, etc.

The Bidder shall be responsible for any internal device wiring not completed in the manufacturer's factory.

The Bidder's electrical installation work shall be in accordance with the Bidder's or manufacturer's drawings, as reviewed and approved by the Owner. They shall be in compliance with equipment manufacturer's instruction manuals and recommendations where applicable and such detailed instructions as may be submitted by the Owner. If for any reason it is necessary to deviate therefrom, the Bidder shall obtain permission from the Owner.

Electrical items are shown on the Drawings in their approximate locations. To avoid interferences or obstructions with other equipment, piping, duct runs, culverts, etc., it shall be the responsibility of the Bidder to check for interferences in the area before installation. If interferences or obstructions to such items occur, either prior to or subsequent to their installation, the Bidder shall relocate such items at no cost to the Owner.

4.4.2. INSTALLATION

4.4.2.1 General

The installation of equipment generally consists of off-loading, receiving the equipment, moving it to the storage or protected area and placing it in the final installation location, unpacking, assembling and setting it in place.

All work shall be done in an orderly and workmanlike manner, shall present a neat appearance when completed, and shall conform to the best modern practice for the class of work notwithstanding any omission in the drawings and the Specifications. It shall be the responsibility of the Bidder to coordinate his installation activities with the material delivery schedule of others and the construction program so that a smooth and harmonious schedule may be maintained. It shall be the responsibility of the Bidder to coordinate his installation activities with others.

The Bidder shall supply and install temporary protective covers, such as plywood or polyethylene sheeting, over equipment which might be damaged by falling materials, dripping water, dust, etc. to the satisfaction of the Owner.

The words “erect”, “install” or “installation” shall include the Work, but not be limited to, the following:

- (1) The setting, plumbing, aligning, shimming, grouting and anchoring of motors and other electrical equipment on foundations.
- (2) Placing of all expansion anchors, mounting bolts, base channels, cable clamps and supports, floor plates, cable and conduit fittings, grouting, etc. as required at the equipment.
- (3) Making of any internal connections on equipment that were omitted for shipment and the installation of necessary jumpers and connections to local junction boxes as defined in the Technical Specifications.
- (4) Mounting and connecting of items shipped separately and removing and replacing details for convenient handling.
- (5) Cleaning and checking of electrical equipment and connections and touching up of damaged surfaces.
- (6) Fabricating and installing steel details for mounting fixtures, conduit, control cabinets and similar items to the satisfaction of the Owner.
- (7) Correction of errors in factory-assembled equipment and other manufactured items.

The Bidder shall supply miscellaneous materials as approved by the Owner for use in making electrical connections and for mounting equipment, as required by the Specifications. These materials generally include, but are not limited to, the following:

- (1) All equipment and materials required for the complete permanent lighting installation including the 230 V AC receptacles and those in hazardous area.
- (2) Temporary construction materials required for the installation and erection of equipment necessary to do the Work. The temporary power and light system shall be included.
- (3) Tapes of all types, solder, flux, Sil-Fos strips, welding rods, welding gases, splicing materials, cable pulling compounds, sealing compounds, varnishes, paints, gasoline, cleaning fluids, sandpaper, emery cloth, rags, twine, oil filter paper, etc.
- (4) Anchors, machine bolts, washers, nuts, screws, etc. (all corrosion proof).
- (5) All structural supports, frames and brackets required for power, control, and instrumentation and lighting conduits and for lighting fixtures and other miscellaneous electrical devices.
- (6) All required conduits, fittings, racks, trays and raceways.
- (7) All steel and plastic electrical ducts and conduits with fittings, supports and junction and pull boxes.
- (8) All terminal lugs for cables and wires, cable supports, stress cones, potheads, cable tags and other necessary cable and wire terminating materials.
- (9) All copper bars, copper cables, fittings and welding and brazing materials for grounding.
- (10) Materials required for buried cable installations, as specified.
- (11) Materials for fireproof cables in manholes and other locations.

All openings in walls (including metal siding) required for the mounting, supporting, or servicing of electrical equipment supplied by or supplied to the Bidder for erection shall be the responsibility of the Bidder. The Bidder shall repair any excessive physical damage in providing openings and shall fireproof and/or weatherproof any openings provided.

The Bidder shall not support his equipment by attaching to grating or piping.

The Bidder shall provide, where necessary, cut-outs for mounting and installing electrical items such as grounding connections, conduits, junction boxes, control devices, local control panels, switches, etc.

The Bidder shall be responsible for the care, safety and maintenance of all the materials until acceptance of the installed materials by the Owner. The Owner will not pay for removal or replacement of damaged or faulty material after it has been installed or if damaged prior to installation.

4.4.2.2 Control Panels and Cubicles

Control panels shall be installed in such a manner that the front of each panel is easily accessible and a minimum of 1.5 meters away from piping, steelwork, cable tray, equipment, etc. There shall be sufficient clearance at the rear of the panels to allow easy access for maintenance, with the door(s) open. There shall be sufficient clearance above the panels to allow easy installation of external piping, tubing and cables.

Erection requirements for the Substation Control Panels are as follows:

- (1) The control panels for the diameters in the Bidder scope shall be placed in the existing control room and connected to the existing panels.
- (2) The new metering panel, therefore, shall be installed close to the existing metering panel, any necessary rearrangement shall be included in the scope of work.

Control panels located outdoors which may be exposed to direct sunlight shall be provided with suitable sun shields.

Control panels smaller than 1,800 mm high shall be mounted on steel support brackets rigidly anchored to the wall, structural steel or floor. The control panels shall be rigidly fastened to the support brackets.

Unless otherwise indicated, miscellaneous electrical equipment mounted to masonry shall be spaced at least six (6) mm from mounting surface by means of washers placed over mounting bolts and anchors. Equipment mounted on steel structures shall be fastened with suitable steel mounting brackets welded to the steel mounting surface. Mounting brackets may be fabricated from structural steel bars, angles, etc., or from channel strut components.

All free standing control panels located outside the electric equipment and switchgear rooms shall be mounted on eighty (80) mm high (from finish floor) concrete pads.

Control panels located in areas subjected to vibration from operating equipment, such as vibrating feeders, shall be provided with shock mounts to protect instruments and devices from damage.

Erection requirements for AC/DC system, including battery, charger and AC/DC distribution panels, are as follows:

- New 220 V batteries if required shall be supplied and installed in existing battery room. The space in the battery room shall be adjusted.

4.4.2.3 General Electrical Requirements

All cables, conduits, fittings, junction boxes and other electrical equipment shall be furnished and installed in accordance with Technical Specification E-001.

4.4.2.4 Conduits, Fittings and Cable Tray

The conduits, fittings and cable tray system installation shall be in accordance with Technical Specifications E-231 and E-232.

4.4.2.5 Cable

All power, control, instrumentation and thermocouple cables shall be installed as specified or required by the application in accordance with Technical Specification E-230.

4.4.2.6 Lighting and Communication

Lighting luminaries and accessories, and communication equipment shall be installed in accordance with Technical Specifications E-270 and E-271 respectively.

4.4.3. CHECKING, TESTING AND COMMISSIONING

4.4.3.1 General

All tests required to ensure a satisfactory installation, adjustment, operation and performance of all electrical equipment and materials erected and installed under and in accordance with the detailed Specifications, the latest industry Standards, and the equipment manufacturer's recommendations and drawings shall be performed in conjunction with the Owner's and manufacturer's field engineers. The Owner will coordinate and participate in equipment checkout, testing and commissioning.

The Bidder shall provide qualified engineer(s) and electrical technicians who shall be responsible for coordinating, supervising and performing all checking and testing work deemed to be required of the Bidder by the Owner. Copies of each test report or measurement shall be furnished to the Owner.

The wiring of control circuits and the control equipment itself shall be tested with normal operating voltages applied to the control circuits but without a power connection to the equipment. During these tests the limit switches and other control devices normally operated by driving machinery shall be operated manually to simulate normal operation conditions through all specified sequences of operation. After the individual items of wiring and control equipment have been tested and proven satisfactory under simulated conditions outlined herein, the equipment shall be operated under power, and limit switches, relays and control devices rechecked and adjusted as required for proper operation. The circuits, machines and equipment shall be operated a sufficient number of times under varying conditions to demonstrate that all

items are correctly installed and adjusted and will perform the required operation for all conditions.

Construction testing and commissioning responsibilities must be established on the Site and shall include as a minimum:

- (1) Extent of total testing to be done.
- (2) Frequency of making records and the format to be used.
- (3) Extent of testing to be performed by the construction forces and/or testing sub Bidder.
- (4) Extent of testing to be approved by the Owner.
- (5) Individuals in whom final responsibility and authority is vested.
- (6) Procedures for obtaining clearances and safety tags for electrical equipment and circuits (see Part 4, Sub subsection 4.4, Clause 4.2).
- (7) Distribution of test reports and records.

The Bidder shall have equipment specifications and bills of material available for reference in connection with the testing.

4.4.3.2 Test Equipment

The Bidder shall furnish all electrical test equipment, gauges, instruments and miscellaneous equipment, to perform all work required, including any specialized test equipment indicated in the Technical Specifications which are required to perform specialized site tests. Where a manufacturer's name is provided, it is understood that functionally equal equipment may be used. The following is a representative list of test equipment used to perform various test which may be required:

- (1) High Voltage Test Equipment
- (2) Protective Relay Test Equipment
- (3) Circuit Breaker Test Equipment

4.4.3.3 Preliminary Testing of Electrical Equipment at Site

When electrical equipment is received in advance of its installation, it shall be placed in safe storage. Electrical equipment which is not designed for damp or outdoor locations shall be stored indoors where ventilation is good and moisture buildup is limited. Temporary connections shall be made to the heaters in all switchgears, motors, generators, panels, etc. to prevent moisture from damaging the equipment.

4.4.3.4 Individual Equipment Tests

Manufacturer's instruction books shall be carefully read prior to the start of any test or energizing of any equipment. The latest industry Standards, the Technical Specifications, and all manufacturer's recommendations shall be followed to establish the testing and commissioning requirements for each piece of equipment.

The following is the list of the minimum recommended test for important equipment.

4.4.3.4.1 General

All electrical equipment and cables, where applicable, shall be subjected to comprehensive insulation tests at an appropriate voltage level in accordance with relevant IEC or approved standards.

	Test Voltage - Volts DC		
	CABLES		SWITCHGEAR
	Between Conductors	Conductor to Sheath	
AUXILIARY CIRCUITS			
Control, metering protection and other auxiliary circuits up to 110 V AC and DC	500V Megger	500V Megger	500V Megger
Above 110 V AC	1000V Megger	1000V Megger	1000V Megger
POWER CIRCUITS			
Up to 110 V	500V Megger	500V Megger	500V Megger
Exceeding 110 and up to 415 V	1000V Megger	1000V Megger	1000V Megger
3,300 V	10,000	7,000	7,500
6,600 V	20,000	15,000	15,000
11,000 V	34,000 (belted only)	25,000	25,000

	Test Voltage - Volts DC		
	CABLES		SWITCHGEAR
	Between Conductors	Conductor to Sheath	
AUXILIARY CIRCUITS			
15,000 V	-	37,000	32,000
22,000 V	-	50,000	45,000
33,000 V	-	75,000	66,000

During these tests the following precautions shall be observed:

- (1) Voltage transformers to be isolated.
- (2) Current transformers secondaries to be short circuited and earthed.
- (3) Cables to be isolated.
- (4) Circuit breaker frames to be earthed.

Further details of site testing are also provided in Technical Specifications for the individual equipment in Section 3.

4.4.3.4.2 Generators, Exciters and AVR

Tests shall be carried out in accordance with IEC 60034 and shall include, the following:

- (1) Insulation resistance and polarization index test of generator and exciter to be measured and acceptance value agreed with the Owner.
- (2) Vibration test at 100% synchronous speed of the complete unit.
- (3) Overspeed test at 110% rated speed.
- (4) Voltage balance and phase sequence tests.
- (5) Temperature rise test at MCR.
- (6) Response of AVR and frequency with sudden load application and shedding.
- (7) Open circuit test with step response of voltage reference $\pm 10\%$ and $\pm 2\%$ of reference voltage.
- (8) Full load rejection test. Following full load rejection at rated power factor the generator voltage should have recovered within $\pm 2\%$ of nominal voltage in two (2) seconds.
- (9) Not applicable

- (10) The generator stator, field, exciter, etc., shall be meggered and tested in accordance with the applicable requirements of Clause 3.4.10.
- (11) Check exciter field breaker for proper functioning and to determine if installation of the field discharge resistor is correct.
- (12) Check condition and operation of exciter, silicon diode power rectifier unit, and voltage regulator unit and excitation transformers in accordance with manufacturer's instructions.
- (13) Check voltage transformer, fuses, polarity and turns ratio. Clean all bushings and insulators.
- (14) Measure the resistance of neutral grounding resistor.
- (15) Phase rotation of main generator shall be checked and proper synchroscope operation shall be confirmed.
- (16) Check control of governor and speed changer travel time from control board.
- (17) Computer indicator readings should be compared with temperatures calculated from resistance measurements taken on the RTDs at the machine terminal block with external cables disconnected.

4.4.3.4.3 Power Transformers

High voltage tests shall be carried out on Site following erection. The following tests shall also be carried out:

- (1) Insulation resistance test between core and earth.
- (2) Insulation resistance test between windings and to earth.
- (3) Continuity test for on-load tap changers by applying 400 V, 3 phase to high voltage windings and measuring magnetization current.
- (4) Dielectric strength measurement and moisture content test of oil in oil-filled transformers.
- (5) Calibration of temperature instruments.
- (6) Arcing horns shall be checked for correct settings.
- (7) Operation test on all circuit tapping switches and on-load tap change mechanism.
- (8) Nitrogen shall be tested for oxygen content.
- (9) Carefully inspect all internal hardware including terminal boards.
- (10) Check terminal wiring against the wiring drawings.
- (11) Top and bottom oil samples shall be taken and tested for dielectric strength, purity and clarity.
- (12) Perform a turns ratio test on all taps. Verify internal tap mechanism sheets in accordance with all external settings.
- (13) DC overvoltage tests shall be applied where practical for determining insulation values.

- (14) Check proper grounding of the transformer.
- (15) Check all cooling fans for proper rotation prior to energizing. Inspect all pressure gauges for adequate operation. Inspect all radiator and cooler valves for proper position and sealing of packing glands.
- (16) Measure capacitance and power factor of windings.
- (17) Check the operation of all thermal devices, alarm and high temperature trip using values recommended by manufacturer.
- (18) Check pressure relief devices for proper operation.
- (19) Megger transformer windings and all connecting cable or bus work.
- (20) Verify phasing of all transformers by checking the nameplates against the specifications.
- (21) Energize transformer and observe.
- (22) Set inert gas regulator pressure (if applicable).

4.4.3.4.4 Starting Device

The Bidder shall furnish a list of field tests to be performed on this equipment for the Owner's approval. The Owner reserves the right to request additional field tests at no extra cost, if required.

4.4.3.4.5 Current Transformers

Physical check ratio, polarity and phasing on all CTs against the drawings and characteristic curves, including checking points on CT magnetization curve.

Where CTs are used for directional relaying, watt meters, watthour meters or var meters, the following tests shall be made:

- (1) Check for correct polarity and connections.
- (2) High current ratio test by injection of primary current. Current readings shall be taken at relays, meters and instrument test blocks to verify ratio with connected burden.
- (3) Check insulation resistance of all windings and capacitors.
- (4) Check isolation medium sample for dielectric strength, purity and clarify.

A magnetization curve shall be obtained for each current transformer in order to:

- (1) Detect damage in transportation or installation.
- (2) Prove that the correct cores have been wired out to the relevant terminals and

- (3) For high impedance relay schemes, to confirm that correct relay settings have been calculated.

The DC resistance of each current transformer secondary winding including connecting leads shall be measured.

The insulation resistance of all secondary circuits shall be measured at 1000 volt and recorded.

Primary current injection tests shall be done on all current transformers using adequate primary current to prove correct ratio, polarity and for differential protection schemes, to prove the correct relative polarities of all current transformers of each scheme. (Refer to Clause 3.4.15)

4.4.3.4.6 Voltage Transformers

Physical check ratio, polarity and phasing on all VTs against the drawings and characteristic curves.

Where VTs are used for directional relaying, wattmeters, watthour meters or var meters, the following tests shall be made:

- (1) Check for correct polarity and connections.
- (2) Check insulation resistance of all windings and capacitors.
- (3) Check isolation medium sample for dielectric strength, purity and clarify.

The transformer ratio and polarity shall be checked using a primary voltage high enough to give a clearly measurable secondary voltage or by using rated primary voltage and comparison with an already proven voltage transformer. The phasing and phase rotation shall be checked. For three-phase voltage transformer a test shall be done to show that energizing each primary winding produces an output from only the correct phase secondary winding. The residual voltage of any open delta or broken delta winding shall be measured with rated primary voltage applied.

4.4.3.4.7 Medium Voltage and Power Distribution Center (PDC) Switchgears

After complete erection and cabling all field-installed main and control circuits shall be subject to a high voltage tests specified in the applicable approved Standard.

The insulation resistance of all circuits shall be measured before and after the high voltage tests.

All equipment shall be operated to prove that the operating gear safety interlocks and tripping devices, etc., are satisfactory over the auxiliary voltage limits.

For adjacent equipments forming part of the complete switchboard, a continuity test shall be carried out to prove the connections made on site, using a low resistance test set.

- (1) Clean all buses and insulators. Remove fuses or draw out all voltage transformers. Rack all breakers out and perform dielectric and grounding tests on all switchgear buses, bus ducts and breakers.
- (2) Verify that the phasing of incoming power connections, transformers, switchgear and auxiliary buses agree with the phasing diagrams.
- (3) Check alignment of drawout type gear including floor steel, main and auxiliary contacts, shutter position indicators, cell and breaker alignment, etc.
- (4) All control and metering circuit terminal wiring shall be verified and compared to the wiring drawings.
- (5) Rack all breakers into the remote test position, electrically operate from local controls.
- (6) Rack all breakers into the remote test position, electrically operate them with remote controls.
- (7) Test ratios of all current and voltage transformers.
- (8) Test operation of trip elements.
- (9) Verify that all relay settings are in accordance with the recommended settings.

4.4.3.4.8 Isolated Phase Bus Ducts

- (1) Resistance

The resistance of each section of bus bar trunking shall be measured and the value, when corrected to 20 °C, shall not exceed the value stated in the Schedules in this Specification.

- (2) Joint Conductivity

The conductance of each joint shall be measured and shall not be less than an equivalent length of conductor.

- (3) Insulation Resistance

Insulation resistance to earth of each section of bus bar trunking shall be measured.

- (4) High Voltage Test

Each section of bus bar trunking shall be tested with high voltage in accordance with the requirements of the relevant standard.

(5) Air Pressure

Each section of bus bar trunking shall be filled with dry air to twice the normal working pressure stated in the “Bus Ducting” Technical Schedule of this Specification for thirty (30) minutes and all joints tested for air tightness with a soap solution.

4.4.3.4.9 Circuit Breakers, Disconnecting Switches and Substation Equipment

- (1) NA
- (2) Carefully inspect all internal hardware including terminal boards.
- (3) Check terminal wiring against the wiring drawings.
- (4) Check proper grounding.
- (5) Check insulating gas quality.
- (6) Measure capacitance, resistance and power factor of insulation after assembly is complete.
- (7) Check the operation of all switches and other devices used for operating the mechanism and retaining insulation using values recommended by the manufacturer.
- (8) Check pressure relief devices for proper operation.
- (9) Apply all tests as indicated in 3.4.15 for control circuitry herein.

4.4.3.4.10 Motors

The Bidder shall perform megger test for windings before making up cable connections and before applying rated voltage and compare the insulation value to the manufacturer’s recommendations. When no minimum insulation value is provided by the manufacturer, the recommended minimum insulation resistance (Rm) for AC and DC motor armature and field windings can be determined by:

$$R_m = kV + 1$$

Where:

Rm = Recommended minimum insulation resistance in megohms at 40 °C of the entire motor winding.

kV = Rated motor terminal-to-terminal voltage.

The winding insulation resistance is the observed reading corrected to 40°C, obtained by meggering the winding for approximately one (1) minute. This corrected reading should be compared to Rm.

Temperature correction should always be made if the winding is not at 40°C when the insulation resistance test is conducted.

For large motors (greater than 100 kW), the measured insulation values are lower than those recommended record room temperature and humidity, then perform a polarization index (PI) test. Dry out windings when required, by an approved method by application of heat. Verify the winding condition after applying heat, with a PI test. The recommended minimum PI value for AC and DC machines is:

<u>Insulation</u>	<u>Minimum</u>	<u>PI</u>
Class A	1.5	
Class B	2.0	
Class F	2.0	

Note:

For a PI test, use a constant- voltage 500/1000 volt megohm meter. The PI is the ratio of the one (1) minute duration insulation-to-ground reading, to that obtained after ten (10) minutes. This is a useful guide to insulation dryness.

Values of 1.5 for Class A, and 2.0 for class B and F and greater are obtained from clean dry windings. The lower the ratio, the greater leakage part of the test voltage to ground. This test method measures the capability of insulation to maintain a capacitive charge, the more moisture, the faster the reading drops off.

Disconnect motor from driven equipment where necessary, check lubrication, starter or breaker and control circuits.

Check that the motor is free of dirt and dust, rotate rotor manually to determine that it turns freely.

For larger (100 kW and greater) or medium voltage poly-phase motors, determine the phase sequence of the motor leads prior to terminating them with a phase and motor test set by complying with the test set operating instruction.

Test and calibrate any protective relays and verify that they are in accordance with the recommended settings.

Bump motor to check for correct direction of rotation prior to coupling to driven equipment. The following cautions should be understood.

It must be recognized that the number of starts be kept to a minimum since the life of the motor is decreased with each start.

When no starting information is provided on large induction motors (over 100 kW) or other limits imposed by the motor manufacturer, the following starting restrictions are set forth:

- (1) Two (2) starts in succession coasting to rest between starts, with the motor initially at ambient temperature.

- (2) One (1) start with the motor initially at a temperature not exceeding its rated load operating temperature.
- (3) If additional starts are required, none should be made until all conditions affecting overheating are normalized.

For large motors which are provided with an insulated bearing to prevent circulating shaft currents, verify that field installed accessories have not shunted this requirement.

Motors having insulated bearings may be checked for bearing and shaft current by any of the following methods:

- (1) Shaft Magnetization. If both ends of the shaft are magnetized, shaft currents have probably been flowing.

After the motor is first placed into operation, observe it periodically for overheating of the bearings or excessive winding temperature.

Make any necessary special tests, including, but not be limited to, those on the excitation system of synchronous motors, commutation of DC motors and certain tests recommended by the manufacturer.

4.4.3.4.11 Motor Control Centers (MCCs) and AC Distribution Panels

- (1) Check terminal wiring against the wiring drawings.
- (2) Check rating and make settings on all overcurrent protection devices according to the recommended settings.
- (3) Verify that phasing of buses agrees with phasing diagrams.
- (4) Check equipment grounds.
- (5) Check for proper operation of all controls and interlocks with control circuits properly energized and loads disconnected.
- (6) Make high current tripping tests if required on breakers and magnetic switches.
- (7) Make overvoltage tests on buses and breakers.

4.4.3.4.12 Batteries and Chargers

Batteries and chargers shall be required to undergo charge and discharge tests at Site similar to those specified under routine tests on batteries and chargers.

- (1) Check that installation procedures are complied with.
- (2) Check connection and polarity and make an inspection of batteries.
- (3) Verify the liquid level of each cell is normal.

- (4) Record specific gravity of each cell corrected to the designed temperature. If the specific gravity reading is 1.20 or lower based on a sample taken from the top of cell, a more accurate method of averaging three (3) readings, one (1) each from the top, middle and bottom of the cell, should be used.
- (5) Record the no-load voltage of each cell.
- (6) Verify the polarity and connections of charger.
- (7) Check the “float” and “equalizing” charger settings.
- (8) Verify the correct operation of the ground detector.
- (9) If required, give the batteries a refreshing charge before placing them into service.
- (10) Perform an acceptance test.

4.4.3.4.13 DC Distribution Systems

- (1) A visual check shall be made of equipment connections, taping of conductors, operation of breakers, etc.
- (2) Check terminal wiring and polarity of DC buses against the wiring drawings.
- (3) Check trip element ratings of main and distribution breakers or fuses.
- (4) Check apparatus grounds at each equipment location.
- (5) Check operation of ground detectors.
- (6) Check all solid state parts and connections.

4.4.3.4.14 UPS System

Tests shall be carried out to confirm correct functional operation.

4.4.3.4.15 Control, Relay and Metering Panels, Instruments and Protection Devices

This category includes all individual and specialized control, monitoring and relaying equipment located throughout the power plant. Such equipment is too diversified to permit specific testing procedures. The following general procedures will apply under the supervision of the Bidder or manufacturer if applicable.

- (1) Visual check all control, instrument and test switches.
- (2) Check terminal wiring against the wiring drawings.
- (3) Check all pushbutton, control and limit switches against the wiring and schematic drawings.

- (4) Check connections, calibrations and functioning of all instruments and meters.
- (5) Make settings on and test all protective relays. These settings and tests may require special information and shall be made by the Bidder. The final settings, calculations and all supporting information shall be provided to the Owner.
- (6) Adjust all switches, protective and timing relays, etc. contact closing and opening of all devices shall be verified by continuity test to demonstrate correct functioning.
- (7) With control circuit properly energized, check power operation of all control features, interlocks, protective devices, remote transmitting and receiving devices, etc. All electrical controls shall be tested by trial operation to see that each control and interlock function operates according to the schematic diagrams and description of operation.

After complete erection and cabling, all circuits shall be subjected to the high voltage test, specified in the relevant IEC or approved standard and where applicable.

The insulation resistance of all circuits shall be measured before and after any high voltage tests.

All test results shall be recorded and forwarded to the Owner. The test sheets shall be to the approval of the Owner.

For AC secondary injection tests, a substantially sinusoidal test supply shall be used.

The operating and resetting level (current and /or voltage) and timing of all relays shall be measured over an agreed range of settings for all relays.

For directional relays, phase shifting transformers shall be used to determine the maximum torque angle and the boundaries of operation/restraint.

Other relays shall be fully tested in accordance with the manufacturer's recommendations or as advised by the Owner.

All DC elements of protection relays shall be tested for operation at 70% rated voltage.

All DC supplies shall be checked for severity of current in rush when energized by switching on or inserting fuses or links.

All current operated relays shall be tested by injection of primary current to record the actual relay setting and as a final proof of the integrity of all secondary connections.

The stability of all differential schemes shall be checked by injection of primary current as required by the section on current transformer site tests.

Tests on load shall also be done to demonstrate stability and operation of protection relays as required by the Owner.

All tripping, control, alarm and interlocking circuit shall be functionally tested to prove satisfactory and foolproof operation and/or resetting. The functional and safety aspects of all shorting and/or isolation links, fuses and switching devices shall be proved.

The total burdens connected to all voltage transformer circuits shall be measured and recorded.

The continuous current drain of all trip circuit supervision relays shall be measured and shall not be greater than half the minimum current required for tripping. The supervision current shall be measured with the circuit breaker (or other device) both open and closed.

4.4.3.4.16 Annunciators

- (1) With control circuits properly energized, check power operation of all control features, interlock, protective devices, remote transmitting and receiving devices, etc.
- (2) Check the annunciator for correct functioning of its operating sequence and controls.
- (3) Check all lamps in annunciator back-lighted window units, test-bell, cut-off and reset operations.
- (4) Check each alarm point of visual annunciators by closing the initiating contact(s) and observe the annunciator's operations.

4.4.3.4.17 Cathodic Protection Systems

Field test shall be performed on the cathodic protection systems for circulating water system and field piping as required **Clause 5 of Part 4 Section 3, E-260 and E-261.**

4.4.3.4.18 Lighting and Small Power Installation

Checks shall be carried out as follows:

- (1) Insulation resistance checks on all wiring and equipment
- (2) Conductor continuity
- (3) Earth resistivity and earth continuity tests shall be conducted for each final sub-circuit and the completed installations as per Article 250 of NFPA70 (National Electric Code).
- (4) Correct wiring polarity check on lighting switches and socket outlets

- (5) A lighting survey to confirm that the illuminations level is in accordance with the requirements of Technical Specification E-270.

Note: This survey shall be repeated at end of warranty period.

4.4.3.4.19 Regular and Emergency Lighting Systems

- (1) Check apparatus ground at each equipment location.
- (2) Check operation of transfer switch including undervoltage relays, etc.
- (3) Check all interlocks associated with emergency lighting.
- (4) Lighting intensities of various locations shall be checked.
- (5) Overcurrent tripping test shall be made on all main supply and distribution breakers.

4.4.3.4.20 Fire Detection Installation

Each and every alarm initiation devices shall be subjected to a quantity of the substance it can detect to prove correct operation of the device and its associated alarm at the control panel.

<u>Device</u>	<u>Substance</u>
Smoke detector	Smoke
Flame detector	Ultra violet lamp
Heat detector	Hot air
Gas detector	Natural gas
Manual Call Point	Release front cover

Each initiation device cable shall be open circuited to confirm loop fault indication.

Every circuit in the detection system, including within the local fire panel, shall be functionally tested to correct operation.

4.4.3.4.21 Main and Auxiliary Circuit Cables

The following test shall be carried out by the Bidder on the cables after installation. Voltage levels shall be as identified in related Electrical Specifications.

Where short lengths of power cable have been jointed and connected to switchgear or transformers, and cannot be easily disconnected, the high voltage test on such cables may, at the discretion of the Owner, be reduced to those appropriate to the switchgear and transformers.

Where disconnectable link are provided in the equipment to enable the cable to be isolated for testing, the Bidder shall disconnect and reconnect these link free of charge as directed by the Owner.

Where it is necessary for the metal sheath of single-core power or other cable to be insulated from earth, e.g. where insulated glands are used, the Bidder shall demonstrate to the satisfaction of the Owner by suitable tests that the required conditions have been fulfilled.

The continuity and identification of all cables shall be verified by the Bidder to the approval of the Owner.

4.4.3.4.22 Low Voltage Power Cables

- (1) Megger each conductor to ground with all other conductors in the cable grounded. This test should be made before the cable is connected to its equipment.
- (2) All conductors should be meggered to ground after cable installation is completed prior to over potential tests and energization of equipment to which the cable is connected.

4.4.3.4.23 Control and Instrumentation Cables

- (1) Check tightness of terminal contact.
- (2) All wiring shall be verified and compared to wiring drawings for correct connections.

4.4.3.4.24 Communication Equipment

- (1) Check all external connections and functionality.
- (2) Adjust equipment according to manufacturer's instructions.

4.4.3.4.25 Grounding System

Test shall include measurement of ground resistances at the following equipment and structures:

- (1) High voltage circuit breakers and disconnecting switches.
- (2) Gas insulated bus.
- (3) Switchyard structures.
- (4) Generator circuit breakers.
- (5) Generator frame and natural grounds.
- (6) Generator voltage transformers and surge protection.
- (7) Instrument transformer grounds.
- (8) All lightning protection and arrester grounds.
- (9) All motors.

- (10) All power transformer frames and neutral grounds.
- (11) All control panels and consoles.
- (12) All panel-boards and bench-boards.
- (13) All power panels equipment.
- (14) Spot check power and control conduits installed in accordance with applicable installation details.
- (15) Other miscellaneous grounds selected at random in a manner representative of the entire installation.

In addition, the grounding system shall be isolated from all external systems and tests carried out to verify ground resistance of each ground electrode, ground bus continuity and resistance of joints in grounding system.

4.4.3.4.26 Emergency Diesel Engine Generator

- (1) Megger the generator armature, field, etc. following the manufacturer's meggering recommendations.
- (2) Calibrate and test all protective relays.
- (3) Check the generator start and trip circuits.
- (4) If provided, check brushes for correct type, proper seating and proper tension and termination to the brush-holder.
- (5) Check for proper equipment grounds.
- (6) Check the neutral grounding resistor and neutral grounding transformer.
- (7) Verify that the diesel generator phase corresponds to its related switchgear (or equipment) bus in accordance with the phasing diagrams.
- (8) Check acceleration from start of cranking to synchronization. Check speed, voltages, frequency, etc.

4.4.3.5 Test on Systems and Subsystems

The output of control system power supplies shall be checked for correct voltage, frequency, polarity, phase and capacity as applicable. Where "main" and "back up" outputs are provided, the automatic changeover system and alarm contacts shall be verified.

Wherever possible systems and subsystems shall be functionally tested using simulated inputs and with all display and recording devices, keyboards, alarm annunciators and final control elements operating prior to tests in conjunction with associated equipment.

Full functional and performance tests shall be carried out in conjunction with the associated equipment.

The Bidder shall set and test protection relays by using current injection of the circuits and verifying correct operating characteristics of the electrical protective relays.

4.4.4. WORK PRECAUTIONS

4.4.4.1 General

The Bidder shall provide suitable safety tags, warning signs and/or temporary protection barriers around temporary openings, open manholes, removed sections of grating, exposed and energized electrical equipment, test areas, or other hazardous areas or conditions. The barriers should be used and placed in such a manner so that they form a physical block to accidental contact with energized equipment or moving mechanical parts. Typical barriers are rope strung around a test area, wooden horses, and tape placed on the floor. These barriers should be accompanied by warning signs stating that testing is being done, or that a particular hazard is present.

4.4.4.2 Obtaining Clearances

The first step before any testing is done is to obtain the Owner's clearance on the equipment to be tested. The person requesting the clearance of the equipment should witness confirmation of the clearance using an approved test stick. The test should be made by the Owner to show that the equipment to be tested is actually de-energized.

Testing personnel should keep the Owner informed of their activities. The Owner will direct or operate the necessary equipment to grant the proper clearances. The testing personnel should assure themselves that equipment is actually cleared, properly tagged and grounded.

Procedures for tagging of equipment should be documented prior to testing between the Owner, testing personnel and other personnel working in the area. These procedures shall be followed to ensure the safety of all those involved.

The position of testing personnel can be hazardous. They should make maximum use of training and the experience of regular operating personnel. Where trained personnel are not available, testing personnel themselves must assume command, provide instructions and take extraordinary precautions to assure safety for themselves and others.

4.4.4.3 Potential Hazards

It is essential for safety reasons that all trapped charges on bushings, coupling capacitors, transformers, cables and transmission lines be discharged either by closing the installed grounding device or applying temporary grounds. Testing personnel should keep in mind that innocent looking voltage and power transformers may be energized from the low side. Coupling capacitors or current transformers may become open circuited and high voltages produced.

Testing personnel doing work on buses, disconnecting switches, breakers, etc. should normally be protected by three-phase grounds on both sides of the de-energized equipment on which they are working. Many tests require that equipment be energized with normal or test voltage which produces hazards not present for normal maintenance or repair work. This overall situation requires careful planning of work and use of well established procedures which should normally include a solid ground on at least one (1) side of the breaker.

Testing personnel at times may find themselves working on outdoor equipment when a lightning storm may suddenly appear. If any testing or work on energized equipment or circuits must be performed during this time, extra precautions should be utilized.

Test areas should be inspected for any sources of sudden noises such as horns, sirens, etc., that may startle test personnel. They should be aware of any such disturbances that cannot be eliminated.

Deficient tools and equipment usually have an adverse effect on test personnel, which will make it necessary for them to improvise and take unnecessary risks. Tools and test equipment should be in good working order before any testing begins.

Testing personnel should wear proper clothing when working. Accidents have been caused by the wearing of loose clothing. Oil soaked clothing is dangerous on the job. Some other hazards are the use of metal watch bands and finger rings which have contributed to the seriousness of electrical shocks.

Means shall be furnished for testing personnel to reach the top of transformers, switchgear, motor control centers, etc. These means could take the form of scaffolding portable stools, step-ladders, etc. The use of such items should be designed so that they cannot slip along the floor and be non-metallic to reduce the risk of falls by testing personnel.

Even though most people are extremely careful to avoid human errors, testing personnel have more chance for error than in other work, in many cases due to lack of discipline. This category includes such accidents as climbing on live equipment, working too close to energized parts, switching errors, failure to properly use protective equipment such as barriers, rubber gloves, face masks, safety goggles, etc. The failure to stay within the prescribed areas, performing certain operations in correct sequence, or to wait for instructions before starting work are further examples.

Equipment failures have accounted for fatal accidents to personnel involved in equipment testing. In general, any apparatus known or suspected of potential failure should be promptly cleared up and de-energized, if possible, in order to avoid accidents to those who may be close to the testing area. It is a well established practice to insist that all personnel present stand well in the clear when a new piece of equipment is energized for the first time.

4.4.4.4 Precautionary Safety Procedures

In order to perform day-to-day work safely to the selves and others, testing personnel involved should try to observe the following procedures in planning for the job:

- (1) Ensure everyone involved thoroughly understands their assignment before they start. Obtain and take to the job, the necessary prints, relay settings, written instructions, etc.
- (2) Plan the work from beginning to end, in order to safely complete the job. This includes a thorough understanding of the test equipment.
- (3) Make a list of the necessary safety devices, tools, test equipment and materials needed and check to see they are in good condition.
- (4) Before any testing is done, be sure that definite clearances have been obtained and that all parties thoroughly understand just exactly what has been cleared for testing and that it is then made safe for testing before any work starts. Since some testing is done on equipment which has already been in service, it is important that clearance has been requested from the proper authority and that it is done far enough in advance for proper procedure.
- (5) If any problems should come up on the job, use proven methods of analysis instead of improvising methods. Short cuts, if not adequately analyzed, may result in accidents. Varying from the agreed upon plan may result in loss of confidence and induce carelessness.
- (6) Check the work area thoroughly before leaving the job to make certain that nothing has been done that will adversely affect safe operation of equipment or cause someone else to have an accident.
- (7) Be sure that all interested parties have been informed of any changes made in operation of equipment. When new protective equipment has been put in service, all interested parties should be notified. Likewise, any unsafe conditions which cannot be corrected immediately, should be reported to the proper personnel, who are responsible to follow through in correcting such conditions.

4.4.5. WORK QUALITY

The Bidder shall be strictly governed by the specified quality assurance/quality control procedures. If for any reason it is necessary to deviate there from, permission shall be first obtained from the Owner. The Bidder shall perform the electrical work in a manner consistent with recognized good practice for power plant service and in accordance with the Technical Specifications and applicable standards.

(Last page of Sec. 4.4)